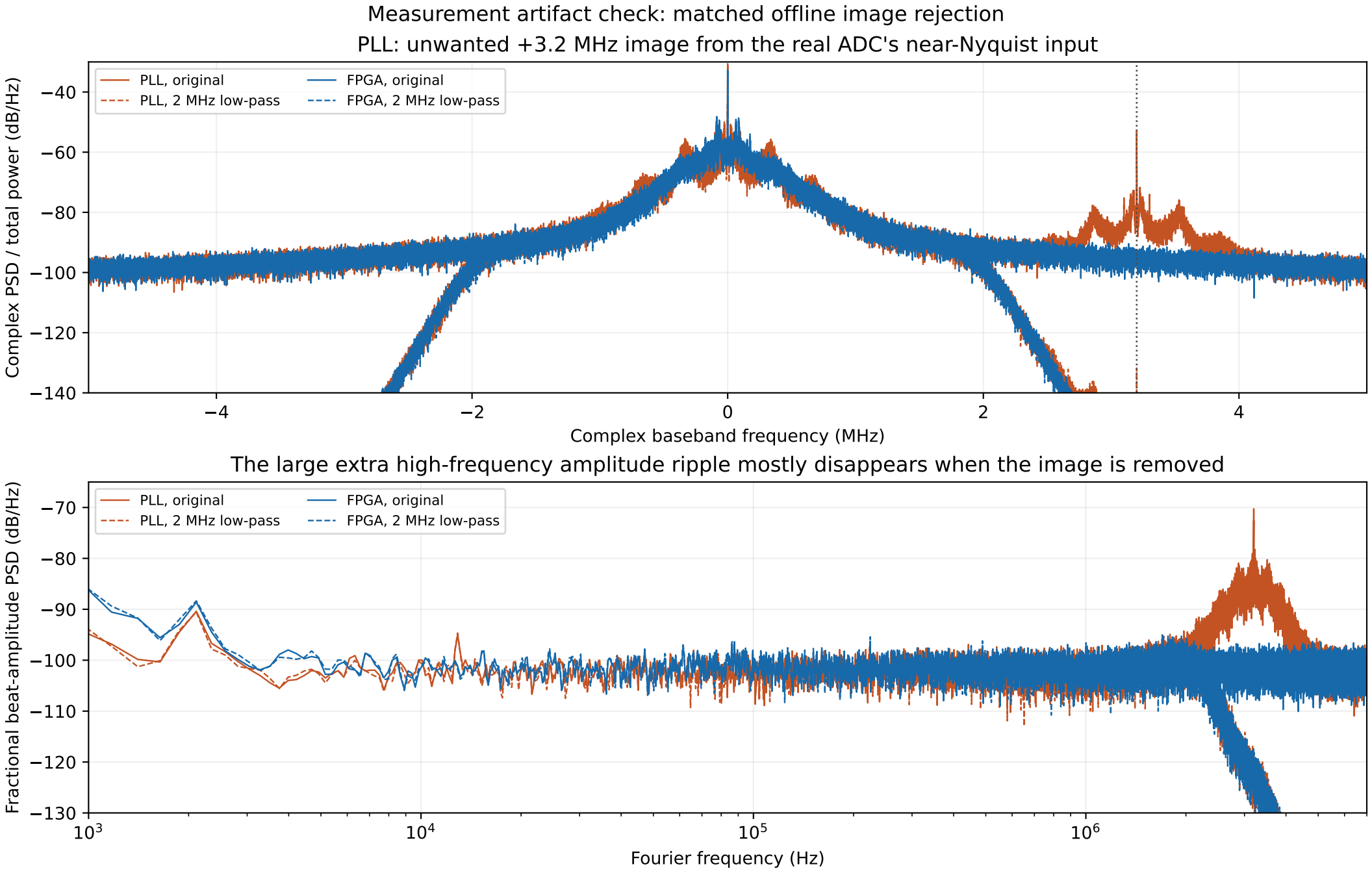

Measurement artifact check: matched offline image rejection
PLL: unwanted +3.2 MHz image from the real ADC's near-Nyquist input
PLL, original
PLL, 2 MHz low-pass
FPGA, original
FPGA, 2 MHz low-pass
Complex PSD / total power (dB/Hz)
−40
−60
−80
−100
−120
−140
−4
−2
0
2
4
Complex baseband frequency (MHz)
The large extra high-frequency amplitude ripple mostly disappears when the image is removed
PLL, original
PLL, 2 MHz low-pass
FPGA, original
FPGA, 2 MHz low-pass
Fractional beat-amplitude PSD (dB/Hz)
−70
−80
−90
−100
−110
−120
−130
10^3
10^4
10^5
10^6
Fourier frequency (Hz)